

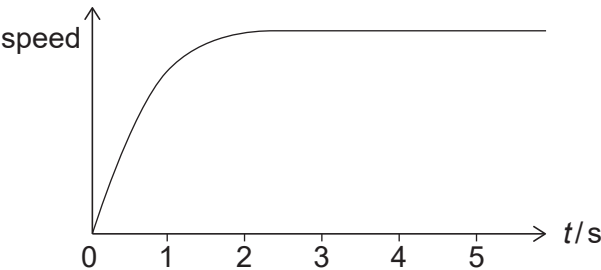
7. A raindrop falls vertically from rest.

(a) State the initial acceleration of the raindrop. [1]

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The graph shows how the speed of the raindrop varies with time t .



(b) Explain, by reference to the vertical forces, how the raindrop reaches a constant speed. [3]

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(Question 7 continued)

- (c) During the first 3.0 s of motion, the raindrop falls a distance of 21 m and reaches a speed of 9.0 m s^{-1} . The mass of the raindrop is 34 mg. The temperature of the raindrop does not change.
- (i) Determine the energy transferred to the air during the first 3.0 s of motion. State your answer to an appropriate number of significant figures. [3]

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- (ii) Describe the energy change that takes place for $t > 3.0 \text{ s}$. [1]

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